

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

REPLY COMMENTS OF THE ADVANCED ENERGY MANAGEMENT ALLIANCE
(AEMA)

I. INTRODUCTION

The Advanced Energy Management Alliance (AEMA)¹ respectfully submits these Reply Comments in response to the Federal Energy Regulatory Commission's ("Commission") October 27, 2025 Notice Inviting Comments regarding the interconnection of large loads. AEMA submits these comments to address the record established during the initial comment period, which includes a wide cross-section of leading industry voices in support of encouraging and rewarding flexible loads, such as data centers.

Review of the over 200 initial comments reveal substantial support—from approximately 112 supportive commenters versus roughly 15 cautionary voices (and 45 neutral commenters)—for the implementation of expedited interconnection pathways and larger interconnection capacities for loads capable of flexibly managing their demand. Specifically, comments filed by transmission owners, technology and data center companies, and solution providers reinforce the need for a technology-neutral framework—such as the "Flexible Load Fast Track" (FLFT) proposed in AEMA's initial comments—that recognizes that load flexibility can be achieved through software and hardware solutions and rewards flexibility based on performance-based criteria.²

In advancing this framework, AEMA urges the Commission to adopt an expressly 'all-of-the-above' approach in which any particular technology (whether storage, thermal resources, on-site generation, or software-defined controls) or combination thereof is encouraged so long as it can verifiably deliver the flexibility attributes that support reliability. While AEMA supports the ability of customers to "bring their own generation" to expedite interconnection, the record demonstrates that this must not be the exclusive pathway. The fastest way to connect new loads that advances affordability, effectively drives investment into U.S. competitiveness and AI

¹ These reply comments are submitted jointly by the Advanced Energy Management Alliance (AEMA) and the following companies: Emerald AI, Lambda, Fluence, Voltus, Camus, ClearPath, Calibrant Energy, and Luminary Strategies (collectively, the 'DC Flexibility Joint Filers,'). This group represents an expansion of the DC Flexibility Joint Filers coalition that submitted initial comments. The DC Flexibility Joint Filers are in broad agreement on the principles and recommendations set forth herein. While the DC Flexibility Joint Filers support the overall thrust of these comments, endorsement does not necessarily extend to every individual statement or characterization.

² Advanced Energy Management Alliance, Comments on Interconnection of Large Loads to the Interstate Transmission System, Docket No. RM26-4-000 (filed Nov 21, 2025). https://elibrary.ferc.gov/eLibrary/docinfo?accession_number=20251121-5374

infrastructure rapidly, and bolsters grid reliability is to harness the full suite of flexibility technologies in a technology-neutral and performance-based way.³

This record confirms that the "Flexible Load Fast Track" proposed by AEMA—and echoed in proposals from other commenters—is not merely a theoretical preference but an urgent economic necessity. However, AEMA must also forcefully rebut the minority of comments from firms and organizations that seek to undercut the value of load flexibility as a valid grid resource.

II. INDUSTRY CONSENSUS FOR FLEXIBILITY AS A "BRIDGE" TO INTERCONNECTION

The record reflects a convergence of opinion among diverse stakeholders that flexible load capabilities should be leveraged to bridge the gap between immediate interconnection needs and the longer timelines required for traditional transmission upgrades, accelerating AI infrastructure investments while also deferring or avoiding upgrades that can raise costs for ratepayers. The record supports the inclusion of energy resources that contribute to flexibility as well as software defined flexibility solutions as valid mechanisms for load management.

- **Constellation Energy Generation, LLC ("Constellation")** states that "interruptible service can serve as a bridge to full service while transmission upgrades are completed" (page 24). Constellation further notes that an "interruptible co-located load does not require the transmission system to be planned, built, or adapted in any way" to meet the full potential demand because the load can be interrupted (page 25).
- **National Grid** provides direct transmission owner support for this approach, stating they are "actively working to implement solutions to make this possible, including through our engagements with companies such as Emerald AI which focus on advanced analytics and flexible demand management" (page 2). National Grid affirms that large loads, "if integrated with flexibility... can help contribute to overall cost efficiency" (page 2).
- **Southwest Power Pool (SPP)** is cited by multiple commenters, including the Arkansas Public Service Commission, for developing the "Conditional High Impact Large Load Service (CHILLS)," a tariff offering designed to allow large loads to connect subject to curtailment or interruption until requisite transmission is available (page 6).
- **Google** similarly recommends that the Commission "prioritize reforms that will expedite the interconnection of facilities that maximize the use of existing infrastructure," specifically for loads that "agree to be curtailable" (page 11).
- **Microsoft** similarly supports expedited study processes for loads that "voluntarily agree to be curtailable," noting this is critical for "accelerating the deployment of critical AI infrastructure" (page 10).
- **The Industrial Consumer Organizations** also propose the XL Connect framework—"an optional federal, fast-lane interconnection process...to enable rapid, transmission-

³ Luminary Strategies (Arushi Sharma Frank), Comments on Accelerating Speed to Power / Winning the Artificial Intelligence Race RFI, 90 Fed. Reg. 45,032. <https://app.halcyon.io/documents/eb341529-2c12-4dd4-baab-f297c6b4e403/attachment>

level interconnection of large electricity loads for customers for whom speed-to-market and/or interconnection certainty is critical, and who can make adequate financial and contractual assurances to the grid" (page 17).

- Various commenters proposed specific mechanisms to operationalize this bridge. **OpenAI** advocates for an "optional curtailable-load pathway" where curtailment obligations allow the transmission provider to "expedite the interconnection request" (page 2).
- As part of the Load Flexibility Parties, **Voltus** and its co-commenters explain, "load flexibility helps balance supply and demand by reducing demand when called upon by the grid operator," underscoring that such flexible capacity is already a proven reliability tool that can be harnessed to facilitate large-load interconnections.
- **Calibrant Energy** highlights their project in the Pacific Northwest where a behind-the-meter energy storage system at a data center "will enable the facility to come online and scale operations years earlier than would be possible with traditional utility upgrades" (page 4) as a real world example to support of their recommendation that load flexibility may "allow a large load to interconnect provisionally while needed infrastructure is built for firm service to create a path for large loads to come online quickly but reliably in the interim" (page 2)

III. DEFINING TECHNOLOGY-NEUTRAL, PERFORMANCE-BASED FLEXIBILITY

The comments filed in this proceeding support a definition of flexibility that is performance-based rather than asset-specific. This performance-based construct ensures that the Final Rule remains genuinely technology-neutral, empowering an all-of-the-above portfolio of solutions rather than presuming that any single class of assets is inherently superior. Stakeholders argue that whether flexibility is achieved through batteries, thermal storage, on-site generation, software orchestration of computational power use, or adjacent controllable resources, the value to the grid remains consistent.

Diverse Technologies for Flexibility: Commenters highlight that flexibility stems from various technologies, all of which should be eligible under a FLFT.

- **Fluence** submits that Battery Energy Storage Systems (BESS) are "uniquely positioned to make large loads curtailable and flexible" and this "minimizes the system impact from interconnection and can minimize the need for expensive and time-consuming network upgrades" (pages 2-3).
- **The Thermal Battery Alliance** notes that thermal energy storage can decouple industrial heat demand from electric demand, explaining that its "operational profile allows these hyper flexible loads to avoid periods of peak scarcity while absorbing energy during low-value hours" (page 2).
- **Emerald AI** notes that "the Commission should adopt a technology-neutral, performance-based definition of flexibility that includes software-defined flexibility as a valid form of 'curtailable and dispatchable' load..." (page 1). Emerald AI adds that this

approach creates clear economic incentives for verifiable flexible loads, "The FLFT, enabled by verifiable software-defined flexibility, creates a clear economic choice: projects that remain inflexible proceed through the standard process and may bear upgrade-related costs, whereas projects that commit to and deliver flexibility can avoid or reduce those upgrades and connect sooner" (page 6).

Transparent and Clearly-Defined Needs: AEMA asserts that transmission providers should offer parallel and expedited study tracks for loads willing to flexibly manage their consumption. AEMA recommends that the interconnection study process be used to define clear, upper-bound limits on the required flexibility (e.g., maximum number of hours per year, hours per event, depth of flexibility, and events per year) needed to mitigate specific system constraints. This recommendation aligns with broader calls for regulatory action: "FERC and state commissions should encourage transmission providers to reform their processes to better utilize voluntary flexible loads, including establishing clear operating parameters (such as frequency, duration, and curtailment notice) and implementing reasonable telemetry and verification requirements to ensure reliable system operation."⁴ This approach is based on established reliability modeling and ensures that data center operators have the operational certainty required to invest in these high-value facilities, achieving both grid needs and economic productivity. This is supported in the record by parties who stressed the need for transparency in order to develop flexibility solutions.

- **Calibrant** states that "the clear identification of system needs will allow customers to develop solutions to address those needs" and "without visibility into what is causing the system to bind, customers will be unable to develop alternative solutions, leading to delayed interconnections, economic loss, and potentially an overbuilt, unnecessarily expensive electric system." (pages 1-2)
- **Critical Loop** recommends "requiring publication of hosting-capacity-style data for load (hourly and seasonal headroom, constraints, expected curtailment windows) and by offering standardized flexible options (limited-load profiles, non-firm service, and joint-study paths for hybrid or adjacent resources)" (page 2)
- **Equinix** claims that to enable "aggregate resource" proposals to resolve or mitigate constraints or violations that would otherwise trigger incremental system costs the approach would "necessitate study transparency requirements that allow large load interconnection customers to understand the specific constraints and resulting incremental energy system investments their energy interconnection might cause." They go on to recommend that "interconnection processes should give large load interconnection customers an opportunity to respond to study results and bring forth solutions that could avoid, defer, or mitigate the need to build upgrades or make other incremental investments." (page 7)

⁴ *Flexible Data Centers: A Faster, More Affordable Path to Power*, Camus, Encoord, & Princeton ZERO Lab, December 2025

IV. STANDARDIZING STUDIES BASED ON NET WITHDRAWAL AND CONTROLLED LIMITS

Multiple commenters argue that interconnection studies must reflect the actual, controlled operating profile of a facility rather than a theoretical maximum withdrawal.

- **Critical Loop** proposes the establishment of a "Large-Load Flexibility Commitment (LLFC)," which they define as "a contractual, tariff-backed commitment by a transmission-connected load to curtail to pre-specified levels" (page 5). They argue transmission providers should "model LLFC parameters explicitly in interconnection studies" rather than defaulting to a static maximum-withdrawal assumption (page 2).
- **Constellation** asserts that "the final large load interconnection rule should reflect the withdrawal rights and different types of service that large loads and hybrid facilities wish to take, rather than requiring every resource to take full network service..." (page 21).
- **Google** supports transmission service based on "actual withdrawal rights" consistent with cost causation principles (page 12).

AEMA agrees and posits that the standardized operator-facility interface referenced in our initial comments (and supported by the capabilities cited by National Grid regarding Emerald AI) provides the necessary "real-time direct telemetry" requested by grid operators.

V. THE COMMISSION MUST REJECT PROPOSALS THAT IGNORE DEMAND FLEXIBILITY OR NARROWLY REQUIRE GENERATION PAIRINGS TO ACCELERATE LOAD CONNECTION

AEMA strongly urges the Commission to reject arguments from the Independent Market Monitor for PJM ("PJM IMM"), the Electric Power Supply Association ("EPSA"), the Illinois Attorney General, and others suggesting that large loads must "bring their own generation" or that demand-side solutions are inferior to generation assets.

A. Matching load with generation should be just one of multiple pathways to dispatchability, flexibility, and curtailability. The PJM IMM's recommendation that "new data center load be required to bring their own new generation" (page 23) and EPSA's call for connecting loads to "designated generation supply" (page 9) ignore the fundamental efficiency of the transmission grid. Mandating that every new large load be paired with new generation forces the construction of redundant capital assets when cost-effective flexibility could otherwise allow that load to be served by the existing system. This approach would unnecessarily inflate the cost of AI infrastructure and add years of supply chain backlog. It would also delay U.S. competitiveness without providing additional reliability benefits that verifiable curtailment can already achieve.

AEMA does not oppose customers electing to pair large loads with new generation where that is the most efficient way to meet their needs; rather, AEMA urges the Commission to treat generation matching as one optional pathway among many to achieve flexibility, not as a mandatory precondition for timely interconnection. In an improved framework, customer-sited generation, software-defined flexibility, storage, and other tools would be evaluated on a

technology-neutral, performance-based basis for how effectively they satisfy reliability and adequacy criteria.

B. Data Center Flexibility is Not "Illusory": AEMA specifically rebuts the PJM IMM's characterization of demand-side options as "illusory" (page 22). Modern AI data centers are not traditional industrial loads relying on manual intervention. As noted by National Grid in the record, today's flexibility is driven by advanced analytics, automated orchestration, and precise telemetry. When a large load commits to a tariff-backed operational limit—monitored in real-time by the grid operator—it provides a reliability service physically equivalent to generation. The capabilities of AI to provide flexible power consumption to relieve grid stress in real time has been rigorously peer-reviewed in the literature, and AI data centers can provide credible, enforceable, and verifiable curtailment.⁵

Accordingly, AEMA urges the Commission to modernize its conception of “demand response” for large loads. For roughly the past two decades, demand response has largely meant that customers reduce load a limited number of hours per year in response to utility or ISO signals, often during peak or emergency conditions. Those signaling tools can now support more frequent, granular adjustments in demand. Large, telemetry-rich computing and industrial facilities increasingly use software, advanced controls, and real-time data to adjust demand continuously, not only during occasional “demand response events.” This type of demand flexibility goes beyond traditional event-based demand response to include price-responsive operations that serve the same core purpose of reducing effective load on the system when needed. Tailoring market access and interconnection procedures to the capability of flexible large loads is crucial for enabling innovative business models to operate economically, support reliability, and minimize system impacts.

C. Expedited Treatment for Flexible Loads is Not Discriminatory: AEMA disagrees with the California Department of Water Resources ("DWR") and Transmission Access Policy Study Group (TAPS) who argue that expedited study treatment for curtailable loads is discriminatory (page 1). It is a fundamental principle of interconnection policy that customers who impose different impacts on the system should be treated differently. A load that is legally and technically bound to curtail during constraints does not require the same network upgrades as a static, inflexible load. Granting a "Flexible Load Fast Track" to the former is not discrimination; it is a recognition of the superior reliability attributes of the resource.

D. Exclusion of Flexibility Would Be Unduly Discriminatory and Contrary to Established Jurisdiction: Finally, AEMA posits that a final rule singling out loads or hybrid facilities that pair matching generation as the sole mechanism for expedited interconnection—while excluding verifiable load flexibility—would be unduly discriminatory. From the perspective of grid physics and reliability, a verifiable, dispatchable reduction in demand is functionally equivalent to an increase in supply. To grant expedited interconnection to a load based on its ability to build a generator, while denying the same treatment to a load capable of equivalent performance, would treat similarly situated resources differently without justification. Furthermore, any hesitation

⁵ Colangelo et al., "AI Data Centres As Grid-Interactive Assets," *Nature Energy*, 2025.
<https://www.nature.com/articles/s41560-025-01927-1>

regarding the Commission’s authority to mandate the inclusion of load flexibility is unfounded. The Commission has a proven record of asserting jurisdiction over demand-side capabilities that affect wholesale rates and reliability, such as but not limited to Order No. 745 (demand response compensation) and Order No. 2222 (distributed energy resource aggregation). Just as the Commission asserted jurisdiction to remove barriers to DERs and demand response to ensure just and reasonable rates, it possesses the clear authority—and the obligation—to ensure that load flexibility from a wide range of resources, from storage to software-defined flexibility, is treated on a level playing field with matching generation in the interconnection process.

VI. CONCLUSION

The record established in this proceeding demonstrates widespread industry agreement that flexible load is essential to resolving interconnection backlogs. Stakeholders from across the sector—spanning National Grid, Constellation, Google, Microsoft, Industrial Consumer Organizations, and others—have identified that expediting interconnection for flexible resources reduces the need for immediate physical infrastructure and aligns with efficient grid planning.

Accordingly, AEMA respectfully requests that the Commission establish a Flexible Load Fast Track (FLFT) and adopt a technology-neutral and performance-based definition of flexibility in its proposed large load interconnection rules.

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On behalf of the Advanced Energy Management Alliance
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